Laboratorio de Microeconomía I

Tarea 2

Centro de Investigación y Docencia Económicas
Maestría en Economía
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ASSUMPTION 1.1 (Jehle & Reny): Properties of the Consumption Set, X

The minimal requirements on the consumption set are

1. $\emptyset \neq X \subseteq \mathbb{R}^n_+$.
2. X *is closed.*
3. X *is convex.*
4. $\mathbf{0} \in X$.

Definition 3.C.1 (Mas-Colell et al.): The preference relation $\succsim$ on X is *continuous* if it is preserved under limits. That is, for any sequence of pairs $\{(x^n, y^n)\}_{n=1}^{\infty}$ with $x^n \succsim y^n$ for all n, $x = \lim_{n\to\infty} x^n$, and $y = \lim_{n\to\infty} y^n$, we have $x \succsim y$.

Example 3.C.1 (Mas-Colell et al.): *The Lexicographic Preference Relation.* Let $X = \mathbb{R}^2_+$. Define:

$$x \succsim y \text{ iff } x_1 > y_1 \text{ or } (x_1 = y_1 \text{ and } x_2 \geq y_2)$$

This is known as the *lexicographic preference relation.*

Exercises

Ex. 1.1 Let $X = \mathbb{R}^2_+$. Verify that X satisfies all five properties required of a consumption set in **Assumption 1.1**.

Ex. 1.2 Let $\succsim$ be a preference relation. Prove the following:

(a) $\succsim \subset \succsim$
(b) $\sim \subset \succsim$
(c) $\succ \cup \sim = \succsim$
(d) $\succ \cap \sim = \emptyset$

Ex. 1.3 Give a proof or convincing argument for each of the following claims.

(a) Neither $\succ$ nor $\sim$ is complete.

(b) For any $\mathbf{x}^1$ and $\mathbf{x}^2$ in X, only *one* of the following holds: $\mathbf{x}^1 \succ \mathbf{x}^2$, or $\mathbf{x}^2 \succ \mathbf{x}^1$, or $\mathbf{x}^1 \sim \mathbf{x}^2$.

Ex. 1.4 Prove that if $\succsim$ is a preference relation, then the relation $\succ$ is transitive and the relation $\sim$ is transitive. Also show that if $\mathbf{x}^1 \sim \mathbf{x}^2 \succsim \mathbf{x}^3$, then $\mathbf{x}^1 \succsim \mathbf{x}^3$.

Ex. 1.5 If $\succsim$ is a preference relation, prove the following: For any $\mathbf{x}^0 \in X$,

(a) $\sim(\mathbf{x}^0) = \succsim(\mathbf{x}^0) \cap \precsim(\mathbf{x}^0)$

(b) $\succsim(\mathbf{x}^0) = \sim(\mathbf{x}^0) \cup \succ(\mathbf{x}^0)$

(c) $\sim(\mathbf{x}^0) \cap \succ(\mathbf{x}^0) = \emptyset$

(d) $\sim(\mathbf{x}^0) \cap \prec(\mathbf{x}^0) = \emptyset$

(e) $\prec(\mathbf{x}^0) \cap \succ(\mathbf{x}^0) = \emptyset$

(f) $\prec(\mathbf{x}^0) \cap \sim(\mathbf{x}^0) \cap \succ(\mathbf{x}^0) = \emptyset$

(g) $\prec(\mathbf{x}^0) \cup \sim(\mathbf{x}^0) \cup \succ(\mathbf{x}^0) = X$

Ex. 3.B.3 Draw a convex preference relation that is locally nonsatiated but is not monotone.

Ex. 3.C.1 Verify that the lexicographic ordering (**Example 3.C.1**) is complete, transitive and strongly monotone.

Ex. 3.C.3 Show that if for every x the upper and lower contour sets $\{y \in \mathbb{R}^L_+ : y \succsim x\}$ and $\{y \in \mathbb{R}^L_+ : x \succsim y\}$ are closed, then $\succsim$ is continuous according to **Definition 3.C.1**.